When can AI discover adsorption kinetics?

Abdulhamid Batayhi

Department of Mechatronics Engineering, Faculty of Mechanical Engineering,
Yıldız Technical University, Istanbul, Türkiye

ORCID 0009-0007-0374-6755

A fixed-bed adsorption experiment reports concentrations and temperatures over time, most often at the column outlet. What an engineer needs from it is the uptake rate law, usually written as the linear driving force (LDF), $\partial q/\partial t = k\,(q^*(c,T) - q)$. Equation-discovery methods offer to find the law itself, not only its coefficient: sparse regression over a library of candidate terms [1, 2] and its weak-form [3, 5], ensemble [6] and exact best-subset [9] variants, symbolic regression [10, 11], and neural networks embedded in the governing equations [12]. For adsorption, one study has done this from column data: Santana et al. trained a neural uptake term on noisy outlet breakthrough curves and read the LDF form back out, with the isotherm given exactly and at one rate and one noise level [12]. Repeated database searches found no other study that discovers an adsorption rate law from column data. Where such a result holds, and where it stops holding, is not known.

Part of the answer is classical. Near local equilibrium the mass-transfer coefficient is poorly determined by column data [13]: finite rates then act only through an apparent dispersion [15], the LDF coefficient can be extracted wrongly from outlet breakthrough [16], and most of the output variability belongs to the isotherm [17, 18]. Yet in the constant-pattern regime a lumped rate coefficient is recovered from breakthrough consistently with theory [19], so any limit must be conditioned on dispersion, isotherm shape, noise and isotherm error. Equation discovery has its own limits. On a single transient trajectory that settles onto a slow manifold, as a breakthrough run does, the candidate terms become nearly collinear and the data barely distinguish them [20–22]; noise induces phase transitions in what can be recovered [23, 56, 57]; and unique recovery needs conditions that non-chaotic dynamics generally lack [66]. These results say that discovery must fail somewhere for column data. None says where, for which methods, or whether it fails where the rate constant itself stops being identifiable.

The question matters most where the law is not the textbook one. In the water-harvesting framework MOF-303 the water diffusion rate passes through a minimum at the steep step of the isotherm, and the LDF approximation is least accurate there [58, 59]; a loading-dependent rate has been fitted by hand [44]. A method that could discover such a law from column data would be useful. Whether one can, at the noise and isotherm accuracy of a real experiment, is what this study tests.

We asked, in a pre-registered computational study, for which regimes the adsorption rate law can

be discovered from data a column experiment can produce, and whether discovery fails at the same boundary as the practical identifiability of the rate constant [47], or earlier. Ground truth came from a validated non-isothermal column model [50], for a Langmuir sorbent and a MOF-303-like step isotherm, over four decades of Damköhler number, three Péclet numbers, four noise levels and four levels of isotherm error, observed through interior probes of concentration, uptake and temperature, through probes without uptake, or at the outlet alone. Methods from each family above were run in their strongest available configuration, and identifiability was judged by profile likelihood on the same observations. The design, thresholds and verdict words were frozen before any method ran on the grid. **[PENDING: one or two sentences stating the corrected verdicts.]**

Results

Both isotherms pass the positive control

At the lowest Damköhler number, with no noise and no isotherm error, discovery and identifiability must both succeed in at least 90 % of the estimation replicates, or that isotherm is withheld from every verdict. Both isotherms passed: the best method recovered the LDF law in 135 of 135 replicates, and k was identifiable in 135 of 135, for the Langmuir and for the step isotherm alike. (As first run, the step isotherm failed this check because of a search-range error in the identifiability code; the correction and both sets of verdicts are given in Supplementary Note 2.)

Discovery fails long before identifiability

H2b compares, in each slice of the design (one isotherm, one Péclet number, one noise level and one isotherm error), the Damköhler number at which the best discovery method falls to 50 % success, $\mathrm{Da}^*_{\mathrm{disc}}$, with the number at which k stops being identifiable in half the replicates, $\mathrm{Da}^*_{\mathrm{ident}}$. Of the 96 slices, 30 had both boundaries on the grid. In all 30 the verdict was "*discovery fails before identifiability*", so H2b's pre-registered overall word is the same (Fig. 2, Table 1). The gap was large everywhere: from 1.6 to 3.4 decades of Da, with a median of 2.3 decades for the Langmuir sorbent (20 slices) and 2.5 for the step isotherm (10 slices). Discovery failed between Da = 0.11 and 7.2, while k stayed identifiable up to Da between 43 and 562. In every one of the 30 slices the best discovery method was the same, exact best-subset selection in the weak form.

Noise moved both boundaries, but not equally. With 0.5 % noise and an exact isotherm, discovery held to Da ≈ 5 to 7 and identifiability to between the two highest levels (Da = 316 and 1000). At 2 % noise the discovery boundary fell to Da ≈ 1, and at 5 % to Da ≈ 0.1 to 0.3, near the bottom of the grid, while the identifiability boundary stayed between Da ≈ 43 and 172. A small error in the isotherm hurt discovery more than noise did: at 0.5 % noise, an isotherm error of 0.5 % moved $\mathrm{Da}^*_{\mathrm{disc}}$ from about 6 to about 0.2, more than a decade, while $\mathrm{Da}^*_{\mathrm{ident}}$ moved from 562 to about 360.

The 66 slices without a verdict are not missing data. In each, one of the two probabilities did not cross 50 % anywhere on the grid, and the pre-registered words for that case are "*no boundary on the grid for one of the two*". **[PENDING from the figure data: how many of these never**

reach 50 % discovery at any Da **(expected at isotherm error $\geq$ 2 %) and how many never fail (expected without noise).**] Many identifiability boundaries fell exactly between two adjacent levels of Da, where every replicate was identifiable at one level and none at the next; such a boundary is reported as lying between those levels (Methods). [**VERIFY from the figure data: all-to-none between adjacent levels for every zero-width interval.**]

The verdict does not depend on how success is scored. When success also required the fitted k to lie within a factor of two of the truth (a pre-registered sensitivity analysis), the overall word and all 30 slice verdicts were unchanged.

No single error ratio predicts the boundary, and errors-in-variables could not be tested

H2a asked whether one dimensionless ratio, the realised driving force against the combined error of the observations and of the isotherm ($R = \delta/\varepsilon_{\mathrm{eff}}$), places the discovery boundary at the same value R^* in every condition. A verdict needs at least two conditions whose success crosses 50 % in R. Only one did (the step isotherm at Pe $\approx$ 19, observed through all probes, with $R^* = 123$, 95 % interval 113 to 135), so H2a has "*no verdict*". In the other eleven conditions, success did not cross 50 % anywhere in the range of R the grid produced. [**PENDING from the figure data: whether those stay above or below 50 %.**]

H2c asked whether an errors-in-variables estimator, which treats the measured isotherm as a regressor with error, moves the discovery boundary by at least a factor of two at a given isotherm error. It needs a discovery boundary on the grid for both that estimator and the best ordinary method in the same slice, at isotherm errors of 0.5 to 5 %. No slice had both, so H2c has "*no verdict*" at every isotherm error, and it is neither supported nor falsified. The reason is the one behind most of the 66 H2b slices without a verdict: at these isotherm errors and 2 % noise, the best methods rarely reached 50 % success at any Da. [**VERIFY with the figure data.**]

From the outlet alone, k stays identifiable; the law is not discovered

Real columns are rarely fitted with probes inside the bed. With concentration and temperature probes but no uptake (O2), uptake must first be reconstructed from the gas-phase mass balance. No discovery method reached a 50 % boundary on O2 in any of the 96 slices. [**PENDING from the figure data: whether O2 discovery fails everywhere or succeeds everywhere; expected to fail, because the reconstruction error concentrates at the front.**] Identifiability, computed on the subgrid where every point of the profile is a full column solve (Pe $\approx$ 94, noise 0.5 and 2 %, isotherm error 0 and 2 %), held to Da $\approx$ 57 to 286 from probes without uptake, and to Da $\approx$ 38 to 172 from the outlet alone (O3). At the same noise and isotherm error, discovery from probes of c, q and T, a far richer observation, fell below 50 % success at Da $\approx$ 1 to 7 without isotherm error and [**VERIFY: never reached it**] with 2 % isotherm error. An outlet breakthrough curve therefore still determines k one to two decades of Da beyond the point where the law stops being discovered from probes.

A loading-dependent rate is found only where the front crosses the step

For the step isotherm the study also simulated a rate that falls to 20 % of its value across the isotherm step (L2), a form motivated by the measurements in MOF-303. This part is exploratory. The state dependence was discovered, by the pre-registered rule, almost only at Da = 1: there the best method succeeded in 77 to 100 % of replicates at every noise level and isotherm error, against at most 58 % at any other level and none at Da ≤ 0.32 without noise. **[PENDING: interpretation check. Expected reason: at low** Da **the loading never reaches the step, and at high** Da **uptake is at equilibrium, so the q dependence is only visible when the front passes the step at intermediate rate. Confirm from the loading ranges before stating it.]**

Methods

Ground truth

Ground truth came from the finite-difference reference solver of the companion study [50]: a one-dimensional, axially dispersed, non-isothermal model of a packed column, verified there against closed-form solutions. Uptake followed one of two laws,

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{1}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{2}$$

where s is the logistic function and θ is the fractional loading at the isotherm's step midpoint, so that under L2 the rate falls to 20 % of k_0 across the step. Two adsorbents were simulated with the same column geometry and flow (Supplementary Table 1): the companion study's favourable Langmuir sorbent, and water on a MOF-303-like framework, a cooperative (Type V) isotherm with its step at $P/P_0 = 0.15$, fed at 30 % relative humidity and 30 °C. The Damköhler number $\mathrm{Da} = k\,t_{\mathrm{stoich}}$ counts kinetic time constants per passage of the adsorption front, with t_{stoich} the stoichiometric time of the column (7200 s for Langmuir and 31 050 s for the step isotherm). Nine levels from 10^{-1} to 10^{3} at half-decade spacing therefore span k from 1.39×10^{-5} to $0.139\,\mathrm{s}^{-1}$ for Langmuir and from 3.22×10^{-6} to $0.0322\,\mathrm{s}^{-1}$ for the step isotherm. Three axial-dispersion coefficients, $D_L \times 0.2$, $\times 1$ and $\times 5$ of the default, completed the grid. They give column Péclet numbers $\mathrm{Pe} = uL/D_L$ of about 470, 94 and 19, and the levels are named by these values below (the code and the pre-registration label them by the D_L multiplier, so its "$\times 0.2$" level is the highest Pe). The grid has 81 solves at 400 cells and 400 stored times, each run until the outlet reached 95 % breakthrough, plus 9 isothermal solves for a control described below. The manifest that records every parameter is public.

Observation models, noise and isotherm error

Every method and every identifiability calculation read the data through one function, so all saw the same corrupted observations. Three observation models were used: interior probes of c, q

and T (O1), interior probes of c and T with q hidden (O2), and the outlet $c(t)$ and $T(t)$ only (O3). O1 and O2 used 20 equally spaced probes; all models used 200 times spread evenly over the solved horizon. Noise was independent Gaussian with standard deviation $\sigma \in \{0, 0.5, 2, 5\}\,\%$ of each channel's range. Methods were never given the true isotherm. They were given a measured one, $q^*_{\mathrm{meas}}(c,T) = q^*(c,T)\,[1 + \varepsilon\, g(c/c_{\mathrm{in}})]$ with $\varepsilon \in \{0, 0.5, 2, 5\}\,\%$, where g is a smooth random function of four Fourier modes on $[0,1]$, scaled to unit root-mean-square and drawn once per replicate. Every random draw was seeded by a hash of (cell, replicate, σ, ε, observation model), so each observation can be regenerated on its own. A replicate, one such observation of one cell, was the unit of analysis; each cell had 60.

Discovery methods

Each method ran in the strongest configuration we could verify on known-answer systems before the pre-registration was frozen, and a method that could not be run that way was reported as such rather than replaced by a weaker one. The candidate library was $\{1, c, q, T, cq, q^2, c^2, q^*_{\mathrm{meas}}\}$; under L1 the true law is $k(q^*_{\mathrm{meas}} - q)$. The methods were sparse regression on a Savitzky–Golay estimate of $\mathrm{d}q/\mathrm{d}t$ (strong form) and on its weak, integral form with compactly supported test functions [3, 4], each with three solvers: sequentially thresholded least squares [1], a bagged ensemble with an inclusion threshold of 0.6 [6], and exact best-subset selection by enumeration with the Bayesian information criterion (M4, the objective of [9]); errors-in-variables best-subset regression given the true error levels (M5, an oracle); implicit sparse regression without any isotherm on an isothermal control (M6, SINDy-PI [8]); symbolic regression (M7, PySR [10]); a universal differential equation with a neural network in the uptake term, trained through the column model and read out by sparse and symbolic regression (M1, the pipeline of [12]); and regression of the state on q-free features, which returns the isotherm where uptake is at equilibrium (M9 [22]). On O2, uptake was first recovered from the gas-phase mass balance, and every O1 method was then run on it. M1 could be trained for at most 50 L-BFGS steps or 2 h per fit on the machine available, so it ran as a declared demonstration: Langmuir, Pe ≈ 94, $\sigma = 2\,\%$, $\varepsilon = 0$, O3, the nine Da levels, one replicate. The best parameters seen during training were kept. As first run, a column solve that exhausted its step budget during training ended the fit with no readout (four of the nine fits); the corrected code keeps the best parameters in that case, and both runs are reported (Supplementary Note 2). A Kolmogorov–Arnold network [51] failed its known-answer test and its cost gate before the freeze and was not run on the grid. Method settings, solver thresholds and their known-answer tests are listed in Supplementary Note 1.

Success

Discovery succeeded when the selected terms were exactly those of the true law, every sign was correct and, for L1, the ratio of the q^*_{meas} and q coefficients was -1 within 10 %. The value of k itself was not required. A method that raised an error counted as a failure; no replicate, cell or output was excluded.

Identifiability

Practical identifiability of k was judged by profile likelihood [47] with the true law known, on the same observations and with the same measured isotherm as the discovery methods: k was practically identifiable when the 95 % profile interval (likelihood-ratio threshold 3.84) lay within $[k/2, 2k]$. On O1 the likelihood summed over all probes the fit of the known law, integrated exactly for piecewise-linear $q^*(t)$ along the observed c and T. On O2 and O3 every point of the profile was a full column solve. The profile searched a coarse logarithmic grid of k and refined it around the interval (41 points per pass on O1, 21 on O2 and O3, as pre-registered). Two corrections to the identifiability profiles were made during the study and are reported in full (Supplementary Note 2). First, on O2 and O3 a measured isotherm with $\varepsilon > 0$ could stall the stiff solver at large trial k for days; each solve was therefore capped at 600 s and a capped solve was scored as a failed solve (amendment A5; two O3 rows had capped solves, none next to its interval). Second, the O1 grid first spanned 10^{-5} to $10\,\mathrm{s}^{-1}$, which excluded the step cells at $\mathrm{Da} = 0.1$; it was widened to 10^{-7} to $10\,\mathrm{s}^{-1}$, a peak on the grid edge was made an error, and all O1 rows were recomputed (amendment A6). Before that recomputation, the O1 refinement was found to use 21 points per pass instead of the pre-registered 41 and was set to 41, so every corrected O1 row uses the declared settings.

Boundaries and verdicts

For each slice the boundary Da^* was the 50 % crossing of an unpenalised logistic regression of replicate outcomes on $\log_{10}\mathrm{Da}$, with a percentile bootstrap interval (2000 resamples within each Da level). No boundary was reported, and no verdict issued, when the outcomes did not cross 50 % inside the tested range; the range was reported instead. Before the freeze the estimator's coverage on planted boundaries was 0.94 for a nominal 0.95, with a bias of +0.012 decade. The best discovery method of each slice was chosen on replicates 0–14 and its boundary estimated on replicates 15–59, which removes the inflation that choosing and estimating on the same replicates causes (0.07 decade in simulation); the identifiability boundary was estimated on the same 45 replicates. M5, which knows the true error levels, took no part in this choice.

Three hypotheses were confirmatory, each with verdict words fixed in advance. H2a: success collapses onto one curve in $R = \delta/\varepsilon_{\mathrm{eff}}$, the driving force relative to the combined error, with the spread of R^* across conditions under half a decade. H2b: on O1, for each slice (σ, ε, isotherm, Pe; 96 in all), the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ was called "*discovery fails before identifiability*" at -0.25 decade or less, "*discovery outlives identifiability*" at $+0.25$ or more, and "*discovery fails with identifiability*" in between; the hypothesis-level verdict was the word held by at least two thirds of the slices with a verdict. H2c: errors-in-variables regression moves the discovery boundary upwards by at least 0.25 decade in at least two thirds of the slices at a given $\varepsilon > 0$. With 45 estimation replicates the H2b rule returned a wrong directional verdict with probability 0.001 when there was no gap, and the correct one with probability 0.965 at a gap of 0.40 decade, the smallest effect of interest. A positive control guarded every verdict: in each isotherm's easiest cells (lowest Da, $\sigma = \varepsilon = 0$, O1) the best method had to succeed and k had to be identifiable in at least 90 % of

replicates, or that isotherm's verdicts were withheld. O2 and O3 were reported descriptively, and H2d, whether the loading-dependent rate of L2 can be discovered, was exploratory.

Pre-registration and amendments

The design was frozen on 11 October 2026 (commit `7dcc2fe`) before any discovery method ran on any grid cell, and every confirmatory driver refused to run unless the pre-registration matched its latest freeze commit. Six dated amendments are appended to it. A1 to A4 were made before the grid ran: 60 replicates split into selection and estimation sets, the smallest effect of interest, the oracle kept out of the pooled verdicts, Pe as a slicing axis, errors counted as failures, the decision to run the grid before peer review, and the M1 demonstration. A5 (the solve cap) was made while the grid ran, and A6 (the O1 grid range) after the verdicts were first computed. Both as-run and corrected verdicts are public, and the paper reports both (Supplementary Note 2).

Exploratory analyses

Three analyses were declared in writing before they ran and are reported as exploratory: M1 trained at the scale a universal differential equation needs (E1), the Kolmogorov–Arnold network on its declared subgrid once the machine was free (E2), and how often an interval that meets $[k/2, 2k]$ misses the true k (E3).

Software and computation

The grid ran on a single computer without a graphics processor, one job at a time, from 12 October to 1 November 2026. Package versions are pinned in the repository.

Use of AI tools

During the preparation of this work the author used a generative AI assistant (Claude, Anthropic) as a tool to assist with drafting and language editing, with writing and checking code, and with checking bibliographic records. The author designed the study, made every scientific decision, reviewed and edited all content, and takes full responsibility for the content of the published article.

Data availability

The ground-truth manifest, every results file, the run log and the as-run and corrected verdicts are public in the study repository and archived on Zenodo (doi:10.5281/zenodo.23325994). The ground-truth fields are regenerated from the manifest by `p2_generate.py`.

Code availability

All code is public at https://github.com/abdulhamidbatayhi123/ai-kinetics-discoverability under the MIT licence and archived with the data. The solver is imported unchanged from the companion study's archive (doi:10.5281/zenodo.23323685).

Author contributions

A.B. is the sole author and conceived and designed the study. The use of AI tools is described in Methods.

Competing interests

The author declares no competing interests.

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